

NETWORKED CONTROL

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NAME/FAMILY NAME..... SOLUTIONS.....
PERSONAL CODE.....
SIGNATURE.....
DATE..... 14/2/'23.....

Notes:

- The solutions must be written in the pages of this leaflet.
- At the end of the exam only this file is accepted, **without any additional page**.
- No notes or books are accepted, neither in paper nor electronic form.
- Clarity and precision will be matter of evaluation.

$$1) \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \end{bmatrix} u$$

$$y = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A) a. \quad p_A(\lambda) = \det(\lambda I - A) = \det \begin{bmatrix} \lambda & -2 \\ -2 & \lambda \end{bmatrix} = \lambda^2 - 4 \Rightarrow \begin{array}{l} \lambda_1 = 2 \\ \lambda_2 = -2 \end{array}$$

MODES: e^{2t}, e^{-2t}
THE SYSTEM IS OPEN-LOOP UNSTABLE

$$b. \quad \text{rank} \begin{bmatrix} \lambda_1 I - A \\ C \end{bmatrix} = \text{rank} \begin{bmatrix} 2 & -2 \\ -2 & 2 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = 2 \Rightarrow \lambda_1 \text{ is observable}$$

$$\text{rank} [\lambda_1 I - A \quad B] = \text{rank} \begin{bmatrix} 2 & -2 & 4 \\ -2 & 2 & 4 \end{bmatrix} = 2 \Rightarrow \lambda_1 \text{ is controllable}$$

$$\text{rank} \begin{bmatrix} \lambda_2 I - A \\ C \end{bmatrix} = \text{rank} \begin{bmatrix} -2 & -2 \\ -2 & -2 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = 1 \Rightarrow \lambda_2 \text{ is not observable}$$

$$\text{rank} [\lambda_2 I - A \quad B] = \text{rank} \begin{bmatrix} -2 & -2 & 4 \\ -2 & -2 & 4 \end{bmatrix} = 1 \Rightarrow \lambda_2 \text{ is not controllable}$$

$$c. \quad u = ky = k \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} x$$

IN CLOSED-LOOP

$$\begin{aligned} \dot{x} &= \left(\begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} k \right) x \\ &= \left(\begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 2k & 2k \\ 2k & 2k \end{bmatrix} \right) x = \begin{bmatrix} 2k & 2+2k \\ 2+2k & 2k \end{bmatrix} x \end{aligned}$$

$$p_{A+BK}(s) = \det \begin{bmatrix} s - 2k & -2(1+k) \\ -2(1+k) & s - 2k \end{bmatrix} = s^2 - 4ks + 4k^2 - 4(k^2 + 1 + 2k) \\ = s^2 - 4ks - 4(1 + 2k)$$

$$\begin{cases} -4K > 0 \\ -4(1+2K) > 0 \end{cases}$$

$$\begin{cases} K < 0 \\ K < -\frac{1}{2} \end{cases} \Rightarrow$$

$$K < -\frac{1}{2}$$

d. There are many ways to respond to question d), e.g. by diagonalization, i.e., computation of eigenvectors, etc.
Simpler ways to respond to d) are listed in the following.

(i) We can show that, for both models, the mode corresponding to eigenvalue $\lambda = -2$ is both unobservable and uncontrollable. Furthermore, we compute that, in both cases, the transfer function is

$$G(s) = \frac{4}{s-2}$$

This proves the equivalence of the two models.

(ii) An alternative way consists of finding the most appropriate transformation that allows to bring the original model in diagonal form, without computing the eigenvectors

$$\begin{aligned} \text{DEFINE } \tilde{x}_1 &= \frac{x_1 + x_2}{2} \\ \tilde{x}_2 &= x_1 - x_2 \end{aligned}$$

$$\begin{cases} \dot{\tilde{x}}_1 = \frac{\dot{x}_1 + \dot{x}_2}{2} = x_2 + 2u + x_1 + 2u = 2\tilde{x}_1 + 4u \\ \dot{\tilde{x}}_2 = \dot{x}_1 - \dot{x}_2 = 2x_2 - 2x_1 = -2\tilde{x}_2 \\ y = \tilde{x}_1 \end{cases}$$

DEFINING $\tilde{x}_1 \rightarrow x_1$ THE SYSTEM CAN BE WRITTEN AS
 $\tilde{x}_2 \rightarrow x_2$

$$\begin{bmatrix} \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}$$

The minimal model is :

$$\dot{x}_1 = 2x_2 + 4u$$

B) WE REFER HERE TO THE MINIMAL SYSTEM REALIZATION ABOVE

$$a. \frac{N_B}{h} \geq 2 \log_e 2 = 2.58$$

$$b. h \leq \frac{2}{2.58} =$$

$$c. h = \log_e \sqrt{2} \text{ s}$$

$$N_B = 2 \rightarrow N = 2^2 = 4$$

$$u_{\min} = 0$$

$$u_{\max} = 2$$

$$\text{SYSTEM DISCRETIZATION: } x_{k+1} = f x_k + g u_k$$

$$f = e^{ah} = e^{2 \log_e \sqrt{2}} = e^{\log_e 2} = 2$$

$$g = \frac{b}{a} (e^{ah} - 1) = \frac{4}{2} (2 - 1) = 2$$

$$x_{k+1} = 2x_k + 2u_k$$

$$\bar{x}_i = \frac{b}{a} \left(-u_{\max} + \frac{(u_{\max} - u_{\min})i}{4} \right)$$



$$\bar{x}_0 = 2(-2) = -4$$

$$\bar{x}_1 = 2 \left(-2 + \frac{2}{4} \right) = -3$$

$$\bar{x}_2 = 2 \left(-2 + \frac{2 \cdot 2}{4} \right) = -2$$

$$\bar{x}_3 = 2 \left(-2 + \frac{2 \cdot 3}{4} \right) = -1$$

$$\bar{x}_4 = 2 \left(-2 + \frac{2 \cdot 4}{4} \right) = 0$$

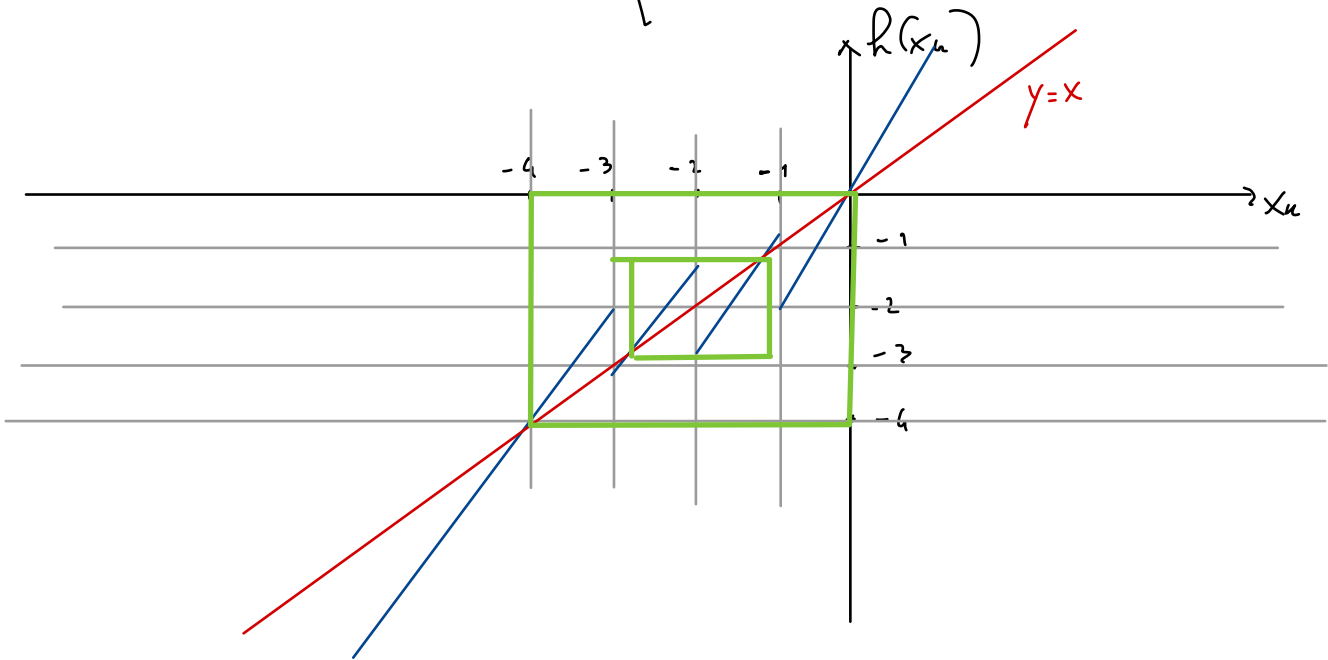
$$\bar{U}_i = U_{\max} - \frac{U_{\max} - U_{\min}}{N-1} i \rightarrow \bar{U}_0 = U_{\max} = 2$$

$$\bar{U}_1 = 2 - \frac{2}{3} = \frac{4}{3}$$

$$\bar{U}_2 = 2 - \frac{4}{3} = \frac{2}{3}$$

$$\bar{U}_3 = U_{\min} = 0$$

IN CLOSED LOOP : $x_{k+1} = h(x_k) = \begin{cases} 2x_k + 4 & \text{if } x_k < -3 \\ 2x_k + \frac{8}{3} & \text{if } -3 < x_k < -2 \\ 2x_k + \frac{4}{3} & \text{if } -2 < x_k < -1 \\ 2x_k & \text{if } x_k > -1 \end{cases}$



USING THE GRAPHICAL METHOD, IF $y_0 \in [-4, 0] \Rightarrow y_k \in [-4, 0]$

$$\text{If } y_0 < -4 \Rightarrow y_k \rightarrow -\infty$$

$$y_0 > 0 \Rightarrow y_k \rightarrow +\infty$$

THE SYSTEM DISPLAYS TWO INVARIANT SET :

• THE LARGEST IS $[\bar{x}_0, \bar{x}_4] = [-4, 0]$, GUARANTEED BY THEORY

• THE SMALLEST IS $[-\frac{8}{3}, -\frac{4}{3}]$. IT CAN BE DEFINED BY

INSPECTING THE GRAPH ABOVE

d. IN CASE $y_0 \in [-4, 0] \Rightarrow y_k = \frac{x_1 + x_2}{2} \in [-4, 0] \quad \forall k$

THE UNOBSERVABLE / UNCONTROLLABLE MODE, ON THE OTHER

HAND, IS SUCH THAT $x_1 - x_2 \rightarrow 0$ AS $t \rightarrow +\infty$

therefore, asymptotically,

$$x_1 \rightarrow \left[-\frac{2}{3}, -\frac{4}{3}\right]$$

$$x_2 \rightarrow \left[-\frac{8}{3}, -\frac{4}{3}\right]$$